

OSINT Reconnaissance and Information Gathering Report

Author

Mythreya KVL

Objective

The objective of this project is to perform Open-Source Intelligence (OSINT) and website reconnaissance using publicly available tools. The assessment demonstrates how information about websites can be collected without direct interaction with target systems.

Scope

The following websites were analyzed:

1. www.nust.edu.pk
2. www.bbc.co.uk
3. www.australia.gov.au

Tools Used

- Google Dorks
- WHOIS Lookup
- DomainTools Reverse IP Lookup
- Wappalyzer
- IdServe
- Wayback Machine
- HTTrack

Methodology

1. Google Dorking

Google Dorks were used to identify websites from different countries.

Queries Used:

- `site:.pk intitle:"index of"`
- `site:.uk filetype:pdf`
- `site:.au inurl:login`

Purpose:

- Discover publicly accessible resources
- Identify websites belonging to specific countries
- Collect target domains for further analysis

2. WHOIS Analysis

WHOIS lookups were conducted to gather:

- Domain registration details
- Registrar information
- Domain creation and expiration dates
- Name servers

Observation: WHOIS records provide valuable information regarding domain ownership and infrastructure.

3. Reverse IP Lookup

Reverse IP Lookup was performed using DomainTools.

Purpose:

- Identify websites hosted on the same server
- Discover related domains
- Understand hosting infrastructure

4. Technology Fingerprinting

Wappalyzer was used to identify:

- Content Management Systems (CMS)
- Web frameworks
- Analytics tools
- Hosting technologies

Observation: Technology fingerprinting helps security professionals understand a website's technology stack.

5. Server Identification

IdServe was used to identify web server information.

Purpose:

- Determine server type
- Analyze HTTP response headers
- Gather infrastructure information

6. Historical Website Analysis

Wayback Machine was used to analyze historical snapshots of websites.

Purpose:

- Review previous versions of websites
- Observe technology and design changes
- Identify historical information exposure

7. Website Mirroring

HTTrack was used to create offline copies of selected websites.

Purpose:

- Preserve website content for analysis
- Perform offline examination
- Study website structure

Findings

- Publicly available sources reveal significant information about websites.
- WHOIS data provides ownership and registration details.
- Reverse IP Lookup identifies domains sharing hosting infrastructure.
- Wappalyzer reveals technologies used by websites.
- Wayback Machine provides historical website information.
- HTTrack enables offline website analysis.

Security Implications

Attackers frequently use OSINT techniques during the reconnaissance phase of a cyber attack.

Organizations should:

- Minimize unnecessary information exposure.
- Regularly audit publicly available information.
- Monitor historical content exposure.
- Review technology disclosure through web applications.

Conclusion

This project demonstrated practical OSINT and reconnaissance techniques using publicly available tools. The exercise highlights the importance of information gathering in cybersecurity assessments and shows how organizations can unintentionally expose useful information through public sources.

Disclaimer

This project was conducted for educational and ethical purposes only. No unauthorized access, exploitation, or malicious activity was performed.